Indian Institute of Science Education and Research, Mohali

Ordinary Differential Equations: MTH401

End Semester Examination: Maximum Marks: 40

Time: 09.00 am - 12.00 noon, November 14, 2025 (1A, AB-II)

Regd. No.

Answer all question with proper justifications

1. Consider the linear system of ODE $X' = AX$: [4+4]
$\begin{bmatrix} x_1'(t) \\ x_2'(t) \end{bmatrix} = \begin{bmatrix} a_{11}(t) & a_{12}(t) \\ a_{21}(t) & a_{22}(t) \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$ where the coefficients a_{ij} are real valued continuous functions on $\mathbb{R}$. Let

$$\Phi(t) = \begin{bmatrix} \phi_{11}(t) & \phi_{12}(t) \\ \phi_{21}(t) & \phi_{22}(t) \end{bmatrix}$$

be a fundamental matrix of the system of ODE $X' = AX$.

   (a) Let $\Psi(t)$ be another fundamental matrix of system $X' = AX$. Show that the matrix valued function $\Phi^{-1}(t)\Psi(t)$ is independent of t.

   (b) For closed and bounded interval $[a, b]$ consider the function $F : [a, b] \times \mathbb{R}^2 \to \mathbb{R}^2$ defined by $F(t, Y) := A(t)Y$. Show that F is uniformly continuous. Prove that F is Lipschitz in the second variable Y for every fixed t.

2. Consider the non-linear system of ODE [3]

$$\begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{bmatrix} nx_1 + x_2 - 2x_1x_2 \\ -2x_1^2 - x_2 + 3x_2^2 \end{bmatrix}.$$

Analyze the nature and stability of the critical point $(0, 0)$ for different values of $n \in \mathbb{Z}$.

3. Show that no non-trivial solution ϕ of the ODE [5]

$$L(y)(t) = y''(t) + (1 - t^2)y(t) = 0$$

vanishes infinitely many times on $\mathbb{R}$.

4. Consider the matrix [5+1+1+1+1]

$$A = \begin{bmatrix} 3 & -4 & 0 \\ 4 & 3 & 0 \\ 0 & 0 & -2 \end{bmatrix}.$$

Find a fundamental matrix of the system of ODE $X'(t) = AX(t)$ and describe the general solutions (real valued). Describe the stable subspace E_s, unstable subspace E_u and center subspace E_c associated with the system of ODE.

5. Consider the interval $I = [0, \pi]$ and functions $p, q \in C^1(I)$ with $p > 0$. Consider the Sturm Liouville problem $L(u) = (pu')' + qu = 0$ with boundary condition $u(0) = 0, \quad u'(\pi) = 0$. [4+4]

   (a) Show that $\int_0^\pi [uL(v) - L(u)v](t)\, dt = 0$ for any $u, v \in C^2(I)$ satisfying the given boundary conditions.

   (b) For $p, q \equiv 1$, construct a Green function G for the above Sturm Liouville problem.

6. Consider the non-linear system of ODE [5]

$$\begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{bmatrix} 2x_1x_2 + x_1^3 \\ -x_1^2 + x_2^5 \end{bmatrix}.$$

Analyze the stability of the critical point $(0, 0)$. Hint: Look for a suitable positive type function of the form $V(x_1, x_2) = a_1x_1^2 + a_2x_2^2$ and use the Liapunov's method.

7. Consider a non-linear system of ODE $Y' = F(Y)$ for which $p \in \mathbb{R}^n$ is a hyperbolic critical point. Consider the associated linear system of ODE $X' = AX$ where $A = DF(p)$. Let

$$N_t(c) = Y(t, c) \text{ and } L_t(c) = X(t, c) = e^{tA}c$$

are the dynamical systems associated with the non-linear and linear system of ODE respectively. Let $h : B_p(\alpha) \to B_p(\beta)$ be a homeomorphism such that $h = L_{-1}hN_1$. Setting the homeomorphism $H = \int_0^1 L_{-s}hN_s ds$ prove that [6]

$$L_tH = HN_t, \ \forall\, t \in [0, 1].$$